

Scaling Book — Section 4 Exercises

Rahul Bir

Q1

$D = 4096$, $F = 4D$, $V = 32,000$, $L = 64$, $NH = D$. Assuming gating-einsum-type MLP and vanilla MHA ($N = K$).

(i)

$$\text{Per-layer params} \approx \underbrace{3DF}_{\text{MLP}} + \underbrace{2D(N+K)H}_{\text{attn}} + \underbrace{D}_{\text{layer norm}}$$

$$\begin{aligned} \text{Total} &= L(3DF + 2D(N+K)H + D) + \underbrace{2DV}_{\substack{\text{embed + unembed} \\ \text{(vocab} \cdot \text{hidden dim size)}}} \\ &= L(3DF + 4DNH + D) + 2DV \\ &= L(3DF + 4D^2 + D) + 2DV \\ &= 64 \cdot (3 \cdot 4096 \cdot 4 \cdot 4096 + 4 \cdot 4096^2 + 4096) + 2 \cdot 4096 \cdot 32000 \\ &\approx 17\text{B params.} \end{aligned}$$

(ii)

$$\frac{\text{attn params}}{\text{total params}} = \frac{4DNH}{4DNH + 3DF + D} \approx \frac{4D^2}{4D^2 + 12D^2} = \frac{1}{4}.$$

(iii)

$$\underbrace{2}_{\substack{\text{keys} \\ \text{and vals}}} \cdot \underbrace{L}_{\substack{\# \text{ of} \\ \text{layers}}} \cdot \underbrace{N}_{\substack{\text{kv heads} \\ \text{per layer}}} \cdot \underbrace{H}_{\substack{\text{hidden dim} \\ \text{per index}}} = 2LD = 2 \cdot 64 \cdot 4096 = 524,288 \text{ bytes} \approx 524 \text{ kB/token.}$$

Q2

$$A[B_x, D_y] \cdot W[D_y, F], \quad \text{mesh} = \{X:4, Y:8, Z:4\}.$$

(i)

Replicated across z -axis so each z -axis “layer” does full matmul.

$$\text{FLOPs}_{\text{Total}} = Z \cdot 2BDF = 2BDFZ.$$

(ii)

Excluding reduction cost from AllReduce:

$$\text{FLOPs/TPU} = \frac{2BDF}{XY}.$$

Q3

$$A[I, J, K, L] \cdot B[I, J, M, N, O] \rightarrow C[K, L, M, N, O].$$

Dimensions I, J are contracting and there are no batching dimensions.

$$\text{FLOPs}_{\text{Total}} = 2IJKLMNO.$$

Q4

(Forward pass only, bf16) *(factor of $\frac{1}{2}$ from self-attn ignored in big- $\mathcal{O}$)*

$$\text{FLOPs} = 2BTSKGH + \mathcal{O}(BTSN) + 2BTSKGH = 4BTSNH + \mathcal{O}(BTSN).$$

Comp order: load Q, K, V from HBM, compute $\text{softmax}(QK^T) \cdot V$ and write back to HBM. (No loading/storing intermediate results from matmuls and softmax.)

$$\begin{aligned} \text{Bytes} &= 2 \cdot BTNH + 2 \cdot BSKH + 2 \cdot BSKH + 2 \cdot BTNH \\ &= 4BTNH + 4BSKH = 4BTKGH + 4BSKH \\ &= 4BKH(TG + S). \end{aligned}$$

$$\text{AI} = \frac{4BTSNH + \mathcal{O}(BTSN)}{4BKH(TG + S)} \approx \frac{4BTSKGH}{4BKH(TG + S)} = \frac{TSG}{TG + S} = \frac{TG}{\frac{TG}{S} + 1}.$$

Prefill / training time ($S = T$):

$$\text{AI} = \frac{TG}{\frac{TG}{T} + 1} = \frac{TG}{G + 1} \quad (\text{almost always compute bound}).$$

Decode (inference-time, $T = 1$, kv-cache full):

$$\text{AI} = \frac{G}{\frac{G}{S} + 1} \Rightarrow \lim_{S \rightarrow \infty} \text{AI} = G.$$

Our # of query groups is small so during decode we're always comm bound.

Q5

$$\text{FLOPs}_{\text{self-attn}} = 4BTSNH + \mathcal{O}(BTSN) \approx 4BTSNH.$$

(Assuming vanilla MHA, $K = N$, $NH = D$.)

$$\text{FLOPs}_{\text{proj}} = 4BTD(N + K)H = 4BTD \cdot (2N) \cdot H = 8BTD^2.$$

$$\begin{aligned} \text{FLOPs}_{\text{self-attn}} = 4BTSN D &\geq \text{FLOPs}_{\text{proj}} = 8BTD^2 \\ \Rightarrow 4BTSN D &\geq 8BTD^2 \Rightarrow 4S \geq 8D \Rightarrow S \geq 2D. \end{aligned}$$

Assuming $D = 4096$:

$$S \geq 8192.$$

Only at large sequence lengths do attn-flops dominate.

Q6

Save:

$$Q[B, T, N, H], K[B, S, K, H], V[B, S, K, H], W_1[B, T, F], W_2[B, T, F], W_3[B, T, D], O[B, T, D].$$

Going backwards through the attn comp graph (figure in section with postnorm):

For $\partial\ell/\partial W_{\text{out}}$, we need to rematerialize:

$$\sigma(W_1) * W_2 \Rightarrow \mathcal{O}(BTF) \text{ FLOPs.}$$

For $\partial\ell/\partial W_{\text{In1}}$ and $\partial\ell/\partial W_{\text{In2}}$, we need to rematerialize:

$$\text{layernorm}_p(x + O[B, T, D]) \Rightarrow \mathcal{O}(BTD) \text{ FLOPs.}$$

For $\partial\ell/\partial W_o$, need:

$$\begin{aligned} \text{softmax}(Q \cdot K) \cdot V &\Rightarrow 2BTSKGH + \mathcal{O}(BTSKG) + 2BTSKGH \text{ FLOPs} \\ &\approx 4BTSNH \text{ FLOPs} \\ &\text{(during training, } T = S) \\ &= 4BT^2NH \text{ FLOPs.} \end{aligned}$$

For $\partial\ell/\partial W_Q$, $\partial\ell/\partial W_K$, $\partial\ell/\partial W_v$, $\partial\ell/\partial W_o$: No rematerialization needed.

$$\Rightarrow \text{FLOPs}_{\text{Total, remat}} = 4BT^2NH + \mathcal{O}(BTD) + \mathcal{O}(BTF) \text{ FLOPs.}$$

Q7

Trained for 2.79M H800 hrs on 14.8T tokens. 37B activated params, w/ FP8 FLOPs, no structured sparsity.

$$\text{FLOPs}_{\text{Total}} = 6 \cdot 37\text{B} \cdot \underbrace{14.8\text{T}}_{\substack{\text{repeated tokens (multiple epochs)} \\ \text{are included in this count}}}$$

$$\begin{aligned} (\text{FLOPs/s})_{\text{utilized}} &= \frac{6 \cdot 37\text{B} \cdot 14.8\text{T}}{2.79\text{M hrs}} = \frac{(6 \cdot 37 \cdot 14.8) \cdot 10^{21}}{2.79 \cdot 10^6 \cdot 60 \cdot 60} \\ &\approx 3.27 \times 10^{14} \text{ FLOPs/s}. \end{aligned}$$

$$(\text{FLOPs/s})_{\substack{\text{peak, H800, FP8} \\ \text{no sparsity}}} \approx 1.5 \times 10^{15} \text{ FLOPs/s}.$$

$$\Rightarrow \frac{3.27 \times 10^{14}}{1.5 \times 10^{15}} = 0.218 \Rightarrow 21.8\% \text{ hardware utilization.}$$

Q8

$X[B, D]$ (flattened), $E \underbrace{[D, F]}_{\text{each expert}}$, k activated experts per token, int8.

$$\begin{aligned} \text{FLOPs} &= 2BDF \cdot k \\ \text{bytes} &= EDF \quad (\text{need to load all experts and then route each token to selected}) \\ &\quad (\text{assuming input is already loaded}) \end{aligned}$$

$$\Rightarrow \underbrace{\frac{2BDFk}{\text{accel FLOPs/s}}}_C \geq \underbrace{\frac{EDF}{\text{accel bw}}}_W \Rightarrow \frac{2Bk}{C} \geq \frac{E}{W} \Rightarrow B \geq \frac{CE}{2kW}.$$

On TPU v5e with $k = 8$, $E = 256$:

$$B \geq \frac{(3.94 \times 10^{14}) \cdot 256}{2 \cdot 8 \cdot 8 \cdot (2 \cdot 10^{11})} = 3940.$$